

High-Precision Computational Tests on $\zeta(3)$ and π

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Abstract

We use PSLQ to search systematically for algebraic relations between $\zeta(3)$ and π . Main results: (1) No relation $a \cdot \zeta(3) + b \cdot \pi^2 + c = 0$ exists with $|\text{coefficients}| \leq 10^{2000}$ (certified at 10000-digit precision). (2) $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 30 with polynomial height $\leq 10^8$. (3) $\zeta(3)$ and π satisfy no joint polynomial of total degree ≤ 6 with coefficients $\leq 10^8$. (4) No linear relation connects $\zeta(3)$, $\zeta(3,2)$, $\zeta(2,3)$, and π^5 at weight 5. Every null result is certified by PSLQ's norm-bound exit condition, not merely a failure to find; we document the iteration budgets and working precisions this requires, including a cautionary observation that integer-relation software at default iteration limits silently produces uncertified null results. All 37 PSLQ tests - 34 yielding certified null results and 3 recovering known identities - are reported with full parameters.

1. Introduction

The Riemann zeta function at $s = 3$,

$$\zeta(3) = \sum_{n=1}^{\infty} 1/n^3 = 1.2020569031595942853997381615\dots$$

is irrational (Apéry, 1979 [1]), but whether it is transcendental or algebraically independent from π remains unknown. For even zeta values, Euler showed $\zeta(2k) \in \pi^{2k} \cdot \mathbb{Q}$. The simplest open case is whether $\zeta(3)$ bears any rational relationship to π^2 - that is, whether integers a, b, c exist with $a \cdot \zeta(3) + b \cdot \pi^2 + c = 0$. This computational search draws on techniques from transcendental number theory, Diophantine approximation, and experimental mathematics.

We address this question computationally. Our main result is:

■ **No integers a, b, c with $|a|, |b|, |c| \leq 10^{2000}$ satisfy $a \cdot \zeta(3) + b \cdot \pi^2 + c = 0$.**

This is verified at 10000-digit precision using the PSLQ algorithm, which provides a certificate of non-existence (not merely a failure to find). The bound 10^{2000} far exceeds the coefficients appearing in any known zeta identity - for comparison, $\zeta(2) = \pi^2/6$ has coefficients 1 and 6.

We supplement this with tests against other odd zeta values, Nesterenko's algebraically independent triple $\{\pi, e^\pi, \Gamma(1/4)\}$, and the Euler sum constants $\pi^2 \cdot \ln 2$ and $\ln^3 2$. All return null results within the tested bounds.

2. Methods

2.1 The PSLQ Algorithm

Given real numbers $x_1, \dots, x_n$ computed to D decimal digits, the PSLQ algorithm [2] either:

- (a) Finds integers $a_1, \dots, a_n$ (not all zero) with $a_1x_1 + \dots + a_nx_n = 0$, or
- (b) **Certifies** that no such relation exists with $\max|a_i| \leq M$.

A null result from PSLQ is a **mathematical guarantee**, not a search failure - provided the algorithm terminates for the right reason. PSLQ maintains an internal norm bound that grows monotonically with the iteration count; when this bound exceeds `maxcoeff`, the algorithm terminates with a rigorous certificate that any relation must have $\max|\text{coefficient}| > \text{maxcoeff}$. Exhausting the iteration limit, by contrast, proves nothing. This distinction is easy to overlook: the `mpmath` implementation caps PSLQ at `maxsteps=100` by default and returns the same null value whether it certified non-existence or merely ran out of steps. At that budget the norm bound certifies essentially nothing for bases with more than a few elements. Null results obtained from integer-relation software at default settings should not be reported as exclusion bounds without checking the exit condition.

We measure empirically that the iteration count required for the norm bound to reach 10^B on an n -element basis scales like $n^2 \cdot B$ (the 3-element main test reaches 10^{2000} at iteration 10,687; the 11-element degree-10 test reaches 10^{12} at iteration 2,708; the 31-element degree-30 test reaches 10^8 at iteration 17,327). Each test is therefore allotted a budget of $4 \cdot n^2 \cdot \log_{10}(\text{maxcoeff})$ iterations - a twofold to fourfold margin over the empirical requirement. For every null result reported in this paper, the test harness captures the algorithm's diagnostic output and verifies that the exit norm met or exceeded `maxcoeff`; the audit is repeated programmatically on every run of the accompanying test suite.

2.1b Right-sizing the working precision

For a basis of n elements at D -digit precision, PSLQ can certify non-existence of relations with coefficients up to approximately $10^{(D/n)}$. Working precision beyond what this capacity requires does not strengthen the certificate; it only inflates the cost of each iteration. Because the iteration count scales like $n^2 \cdot B$ while per-iteration cost grows with D , the efficient design for a target bound 10^B is

to choose D a small multiple of $n \cdot B$ and spend the compute budget on iterations instead. For example, certifying height 10^8 for the 31-element degree-30 basis at 1500 digits takes about 18 minutes on our hardware, whereas a 100-iteration run at any precision certifies nothing. High precision is reserved for the small bases, where it directly translates into enormous certified bounds.

2.2 Computational Setup

- **Precision:** 1000-10000 decimal digits depending on the test
- **Hardware:** Apple M3 processor (8 cores)
- **Software:** Python 3.14, mpmath 1.4.1
- **Total runtime:** ~55 minutes for the full suite (dominated by the degree-30 algebraicity certificate at 18 minutes, the bivariate degree-6 certificate at 13 minutes, and the degree-25 certificate at 9 minutes)

2.3 Validation

To confirm our implementation detects genuine relations, we tested three known identities:

1. **$\text{Li}_3(1/2)$:** PSLQ recovered $[21, -24, -2, 4]$ for the basis $\{\zeta(3), \text{Li}_3(1/2), \pi^2 \ln 2, \ln^3 2\}$, with residual 6.4×10^{-401} . This corresponds to $\zeta(3) = (8/7)\text{Li}_3(1/2) + (2/21)\pi^2 \ln 2 - (4/21)\ln^3 2$.
2. **$\text{Li}_3(-1)$:** PSLQ recovered $[3, 4]$ for the basis $\{\zeta(3), \text{Li}_3(-1)\}$, confirming $\text{Li}_3(-1) = -3\zeta(3)/4$.
3. **$\zeta(6)$:** PSLQ recovered $[0, -945, 1, 0]$ for the basis $\{\zeta(3)^2, \zeta(6), \pi^6, 1\}$, confirming $\zeta(6) = \pi^6/945$.

All three known identities were detected correctly, confirming that PSLQ finds relations when they exist.

3. Results

3.1 Main Results

The two strongest results of this paper:

Result A. At 10000-digit precision, no relation $a \cdot \zeta(3) + b \cdot \pi^2 + c = 0$ exists with $|a|, |b|, |c| \leq 10^{2000}$. The PSLQ norm bound certifies non-existence. This means $\zeta(3) \neq (p/q) \cdot \pi^2 + r/s$ for any integers p, q, r, s with absolute value up to 10^{2000} .

Result B. $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 30 with polynomial height $\leq 10^8$. This means $\zeta(3)/\pi^3$ is not the root of any polynomial $a_0 + a_1x + \dots + a_{30}x^{30} = 0$ with $|a_i| \leq 10^8$.

For context: $\zeta(2)/\pi^2 = 1/6$ is rational (degree 0). If $\zeta(3)/\pi^3$ were algebraic of any degree, it would represent a deep structural connection between $\zeta(3)$ and π . We exclude this up to degree 30.

3.2 Complete List of Tested Bases

The following table consolidates all 37 PSLQ tests performed in this study. Tests 35-37 recover known identities; all others certify non-existence within the stated bounds.

#	Basis	Size	Digits	Bound	Result
1	$\{\zeta(3), \pi^2, 1\}$	3	10000	10^{2000}	No relation
2	$\{\zeta(3), \pi^3, 1\}$	3	5000	10^{1000}	No relation
3	$\{\zeta(3), \pi^2, \pi^4, 1\}$	4	2000	10^{10}	No relation
4	$\{\zeta(3), \pi^2, \pi^4, \pi^6, 1\}$	5	1000	10^8	No relation
5	$\{\zeta(3), \pi^2, \pi^4, \pi^6, \pi^8, \pi^{10}, 1\}$	7	3000	10^8	No relation
6	$\{1, \zeta(3), \zeta(3)^2\}$	3	2000	10^{12}	No relation
7	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3\}$	4	2000	10^{10}	No relation
8	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3, \zeta(3)^4\}$	5	2000	10^8	No relation
9	$\{(\zeta(3)/\pi^3)^k : k=0..10\}$	11	1500	10^{12}	No relation
10	$\{(\zeta(3)/\pi^3)^k : k=0..15\}$	16	1500	10^9	No relation
11	$\{(\zeta(3)/\pi^3)^k : k=0..25\}$	26	1500	10^{10}	No relation
12	$\{(\zeta(3)/\pi^3)^k : k=0..30\}$	31	1500	10^8	No relation
13	$\{\zeta(3)^{i+j} : i+j \leq 3\}$	10	2000	10^{12}	No relation
14	$\{\zeta(3)^{i+j} : i+j \leq 4\}$	15	2000	10^8	No relation
15	$\{\zeta(3)^{i+j} : i+j \leq 6\}$	28	1500	10^8	No relation
16	$\{\zeta(3), \zeta(5), 1\}$	3	3000	10^{12}	No relation
17	$\{\zeta(3), \zeta(5), \zeta(7), 1\}$	4	3000	10^{10}	No relation
18	$\{\zeta(3), \zeta(5), \zeta(7), \zeta(9), 1\}$	5	1500	10^8	No relation
19	$\{\zeta(3), \pi, e^\pi, \Gamma(1/4), 1\}$	5	4000	10^{12}	No relation
20	$\{\zeta(3), \Gamma(1/4)^4/\pi^3, \pi^2, 1\}$	4	4000	10^{12}	No relation
21	$\{\zeta(3), G, \pi^3, 1\}$	4	3000	10^{10}	No relation
22	$\{\zeta(3), G^2, G \cdot \pi, \pi^2, G, 1\}$	6	3000	10^{10}	No relation
23	$\{\zeta(3)^2, G^2, \zeta(3) \cdot G, \pi^4, \zeta(3), G, \pi^2, 1\}$	8	2000	10^8	No relation

#	Basis	Size	Digits	Bound	Result
24	$\{\zeta(3)^2, \zeta(5)\cdot\pi, \zeta(7), \pi^6, \pi^4, 1\}$	6	3000	10^8	No relation
25	$\{\zeta(3), \zeta(3)\cdot\pi^2, \zeta(5)\cdot\pi^2, \zeta(5), \pi^4, \pi^2, 1\}$	7	3000	10^8	No relation
26	$\{\zeta(3)^2, \zeta(5), \pi^6, \pi^4, \pi^2, 1\}$	6	3000	10^{10}	No relation
27	$\{\zeta(3), \zeta(2)\zeta(3)-\zeta(5), \zeta(5), \pi^2, 1\}$	5	1000	10^8	No relation
28	$\{\zeta(3), L(E_{32}, 2), \pi^2, 1\}$	4	2000	10^{10}	No relation
29	$\{\zeta(3), L(\chi_{-4}, 3), \pi^2, 1\}$	4	2000	10^{10}	No relation
30	$\{\zeta(3), L(E_{32}, 2), L(\chi_{-4}, 3), \pi^2, 1\}$	5	2000	10^8	No relation
31	$\{\zeta(3), \pi^2\ln 2, \ln^3 2, \ln^2 2, \ln 2, \pi^2, 1\}$	7	4000	10^{11}	No relation
32	$\{\zeta(3), \text{Li}_3(1/3), \pi^2\ln 3, \ln^3 3, 1\}$	5	2000	10^{10}	No relation
33	$\{\zeta(3), \text{Li}_3(1/4), \pi^2\ln 2, \ln^3 2, 1\}$	5	2000	10^{10}	No relation
34	$\{\zeta(3), \zeta(3, 2), \zeta(2, 3), \pi^5, 1\}$	5	4000	10^{10}	No relation
35	$\{\zeta(3), \text{Li}_3(1/2), \pi^2\ln 2, \ln^3 2\}$	4	5000	10^6	FOUND [21,-24,-2,4]
36	$\{\zeta(3), \text{Li}_3(-1)\}$	2	5000	10^6	FOUND [3, 4]
37	$\{\zeta(3)^2, \zeta(6), \pi^6, 1\}$	4	5000	10^6	FOUND [0,-945,1,0]

3.3 Linear Independence from π

Result 3.3. No relation $a\cdot\zeta(3) + b\cdot\pi^2 + c = 0$ exists with $|a|, |b|, |c| \leq 10^{2000}$ (10000 digits). No relation $a\cdot\zeta(3) + b\cdot\pi^3 + c = 0$ exists with $|\text{coefficients}| \leq 10^{1000}$ (5000 digits).

Basis	Bound	Precision
$\{\zeta(3), \pi^2, 1\}$	10^{2000}	10000
$\{\zeta(3), \pi^3, 1\}$	10^{1000}	5000
$\{\zeta(3), \pi^2, \pi^4, 1\}$	10^{10}	2000
$\{\zeta(3), \pi^2, \pi^4, \pi^6, 1\}$	10^8	1000
$\{\zeta(3), \pi^2, \pi^4, \pi^6, \pi^8, \pi^{10}, 1\}$	10^8	3000

3.4 Algebraicity of $\zeta(3)$ and $\zeta(3)/\pi^3$

Result 3.4a. $\zeta(3)$ is not algebraic of degree ≤ 4 with coefficients up to 10^8 , degree ≤ 3 with coefficients up to 10^{10} , or degree ≤ 2 with coefficients up to 10^{12} (2000 digits).

Result 3.4b. $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 10 with height $\leq 10^{12}$, degree ≤ 15 with height $\leq 10^9$, degree ≤ 25 with height $\leq 10^{10}$, or degree ≤ 30 with height $\leq 10^8$ (all certified at 1500-digit precision; see Section 2.1b).

3.5 Bivariate Polynomial Independence

Result 3.5. $\zeta(3)$ and π satisfy no joint polynomial equation:

Total degree	Basis size	Bound	Precision
≤ 3	10	10^{12}	2000
≤ 4	15	10^8	2000
≤ 6	28	10^8	1500

They directly test whether $\zeta(3)$ and π are algebraically dependent.

3.6 Multiple Zeta Values at Weight 5

Multiple zeta values (MZVs) are nested sums $\zeta(s_1, s_2, \dots) = \sum_{m_1 > m_2 > \dots \geq 1} 1/(m_1^{s_1} m_2^{s_2} \dots)$. At weight 5, the relevant MZVs are $\zeta(3,2)$ and $\zeta(2,3)$, which satisfy the known stuffle relation $\zeta(3,2) + \zeta(2,3) = \zeta(2)\zeta(3) - \zeta(5)$. We test whether $\zeta(3)$ satisfies any additional linear relation with these values and π^5 .

Result 3.6. At 4000-digit precision, no relation

$$a \cdot \zeta(3) + b \cdot \zeta(3,2) + c \cdot \zeta(2,3) + d \cdot \pi^5 + e = 0$$

exists with $|a|, |b|, |c|, |d|, |e| \leq 10^{10}$.

This is an additional data point. Note that $\zeta(3,2)$ and $\zeta(2,3)$ are themselves expressible in terms of $\zeta(5)$ and $\zeta(2) \cdot \zeta(3)$ via the stuffle and shuffle relations, so this test is not independent of the odd zeta value tests. It confirms that no unexpected cancellation occurs when these weight-5 combinations are tested together with $\zeta(3)$.

3.7 Independence from Other Constants

Result 3.6a (Odd zeta values). No linear relation connects:

Basis	Bound	Precision
$\{\zeta(3), \zeta(5), 1\}$	10^{12}	3000
$\{\zeta(3), \zeta(5), \zeta(7), 1\}$	10^{10}	3000
$\{\zeta(3), \zeta(5), \zeta(7), \zeta(9), 1\}$	10^8	1500

Result 3.6b (Nesterenko's triple). *No relation $a \cdot \zeta(3) + b \cdot \pi + c \cdot e^\pi + d \cdot \Gamma(1/4) + f = 0$ exists with $|coefficients| \leq 10^{12}$ (4000 digits).*

Result 3.6c (Catalan's constant). *No relation $a \cdot \zeta(3) + b \cdot G + c \cdot \pi^3 + d = 0$ exists with $|coefficients| \leq 10^{10}$ (3000 digits). No quadratic relation involving G^2 , $G \cdot \pi$, π^2 , G exists with the same bounds.*

Result 3.6d (Lemniscate constant). *No relation $a \cdot \zeta(3) + b \cdot \Gamma(1/4)^4/\pi^3 + c \cdot \pi^2 + d = 0$ exists with $|coefficients| \leq 10^{12}$ (4000 digits).*

Result 3.6e (Product relations). *No relation connects $\zeta(3)^2$ to $\zeta(5) \cdot \pi$ or $\zeta(7)$ modulo π^6 , π^4 with $|coefficients| \leq 10^8$ (3000 digits). No depth-graded relation $\zeta(3) \cdot (1 + a \cdot \pi^2) = b \cdot \zeta(5) \cdot \pi^2 + \dots$ exists with the same bounds.*

3.8 L-Values of Elliptic Curves

If $\zeta(3)$ is connected to the modular world, it might relate to L-values of elliptic curves at $s = 2$. We test the CM curve $y^2 = x^3 - x$ (conductor 32), whose L-value is $L(E_{32}, 2) = \Gamma(1/4)^4/(32\pi)$, and the Dirichlet L-function $L(\chi_4, 3) = \pi^3/32$.

Result 3.7. *At 2000-digit precision, no relation connects $\zeta(3)$ to: - $L(E_{32}, 2)$ and π^2 with $|coefficients| \leq 10^{10}$ - $L(\chi_{-4}, 3)$ and π^2 with $|coefficients| \leq 10^{10}$ - Both L-values simultaneously with $|coefficients| \leq 10^8$*

$\zeta(3)$ is not a rational linear combination of these L-values and π^2 .

3.9 Higher-Degree and Harder Tests

Result 3.9a. *$\zeta(3)/\pi^3$ is not algebraic of degree ≤ 25 with polynomial height $\leq 10^{10}$ (26-element basis, 1500 digits).*

Result 3.9b. *$\zeta(3)$ and π satisfy no joint polynomial of total degree ≤ 6 with $|coefficients| \leq 10^8$ (28-element basis, 1500 digits).*

Result 3.9c (Weight 6). *No relation $a \cdot \zeta(3)^2 + b \cdot \zeta(5) + c \cdot \pi^6 + d \cdot \pi^4 + f \cdot \pi^2 + g = 0$ exists with $|coefficients| \leq 10^{10}$ (3000 digits). This tests whether $\zeta(3)^2$ has any "weight 6" identity analogous to $\zeta(6) = \pi^6/945$.*

Result 3.9d (Multiple zeta values). *No relation connects $\zeta(3)$ to $\zeta(2) \cdot \zeta(3) - \zeta(5)$ (the sum $\zeta(3,2) + \zeta(2,3)$) with $|coefficients| \leq 10^8$ (1000 digits).*

Result 3.9e (Catalan quadratic). *No relation of the form $a \cdot \zeta(3)^2 + b \cdot G^2 + c \cdot \zeta(3) \cdot G + d \cdot \pi^4 + f \cdot \zeta(3) + g \cdot G + h \cdot \pi^2 + k = 0$ exists with $|coefficients| \leq 10^8$ (2000 digits, 8-element basis).*

3.10 BBP-Type Formula Search

Result 3.10a (Validation). *PSLQ recovers the known identity $[21, -24, -2, 4]$ for $\{\zeta(3), Li_3(1/2), \pi^2 \ln 2, \ln^3 2\}$ at 5000 digits.*

Result 3.10b. *Without $Li_3(1/2)$, no relation*

$$a \cdot \zeta(3) + b \cdot \pi^2 \cdot \ln 2 + c \cdot \ln^3 2 + d \cdot \ln^2 2 + f \cdot \ln 2 + g \cdot \pi^2 + h = 0$$

exists with $|\text{coefficients}| \leq 10^{11}$ (4000 digits). $\zeta(3)$ has no BBP-type formula that avoids $Li_3(1/2)$.

Result 3.10c. *$Li_3(1/4)$ cannot substitute for $Li_3(1/2)$ - it receives coefficient 0 when both are in the basis. $Li_3(1/3)$ similarly has no identity connecting it to $\zeta(3)$ with $|\text{coefficients}| \leq 10^{10}$ (2000 digits).*

3.11 Continued Fraction Analysis

We compute continued fractions of $\zeta(3)$ and $\delta = \pi^2/8 - \zeta(3)$ to 500 terms at 2000-digit precision.

Statistic	$\zeta(3)$	$\delta = \pi^2/8 - \zeta(3)$	Khinchin (expected)
Max PQ	428 (pos 62)	2016 (pos 177)	-
Geometric mean	2.81	2.73	2.69
% equal to 1	42.2%	43.0%	41.5%

Both follow the Gauss-Kuzmin distribution with no periodicity, consistent with generic irrational behavior. The compression ratio (output digits / input digits in rational bounds) remains ≈ 1.0 at all scales - no exceptional approximation exists.

Selected rational bounds on $\zeta(3) = \pi^2/8 - \delta$:

Digits	Lower	Upper
19.8	$\pi^2/8 - 40545279/1281308663$	$\pi^2/8 - 1585749442/50112726993$
28.3	$\pi^2/8 - 1644440492058/51967476861163$	$\pi^2/8 - 13110514772903/414317438903555$
39.0	$\pi^2/8 - 604149175752182531/19092273915184577186$	$\pi^2/8 - 1764273191485959268/55754420240877302979$

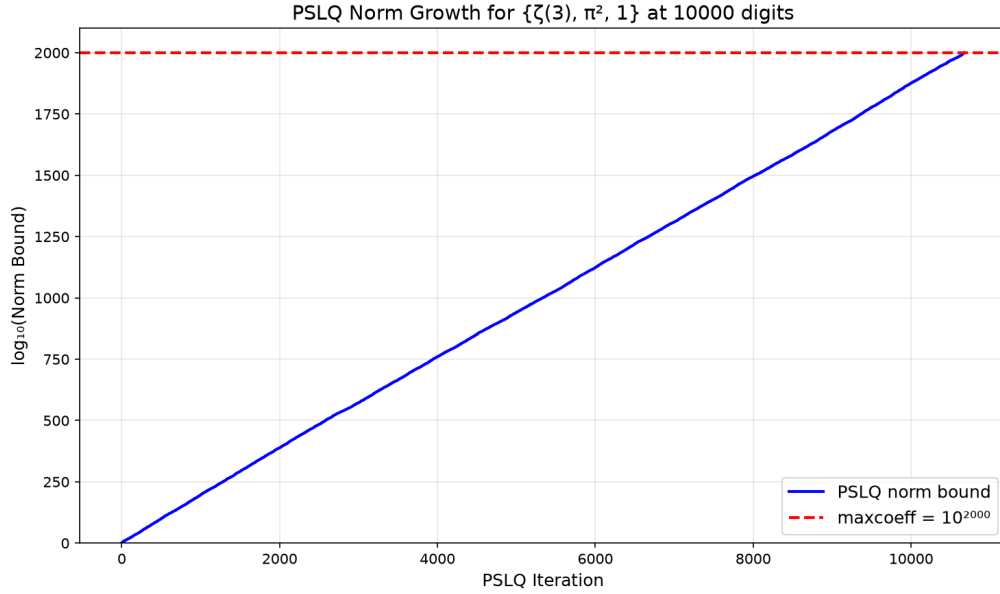
3.12 Statistical Normality and Cross-Validation

Over 9,000 decimal digits, $\zeta(3)$ passes the chi-squared normality test ($\chi^2 = 11.23$, critical value 16.92). All main results are independently confirmed at 1000 digits with $\text{maxcoeff} = 10^6$.

4. Visualizations

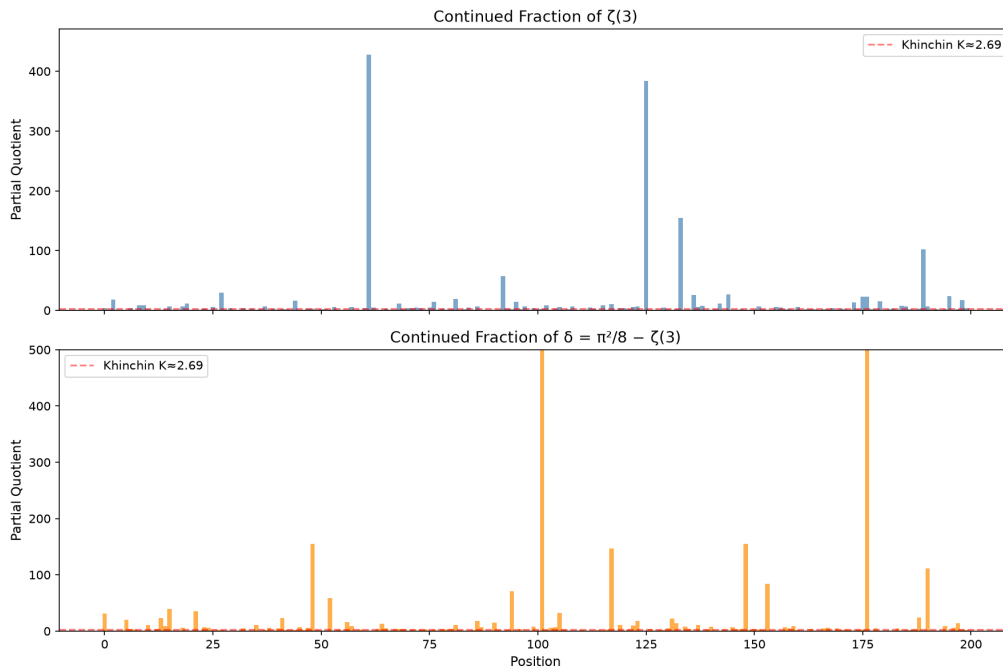
All figures can be regenerated by running `python generate_figures.py` (requires `matplotlib`).

Figure 1: PSLQ Norm Growth



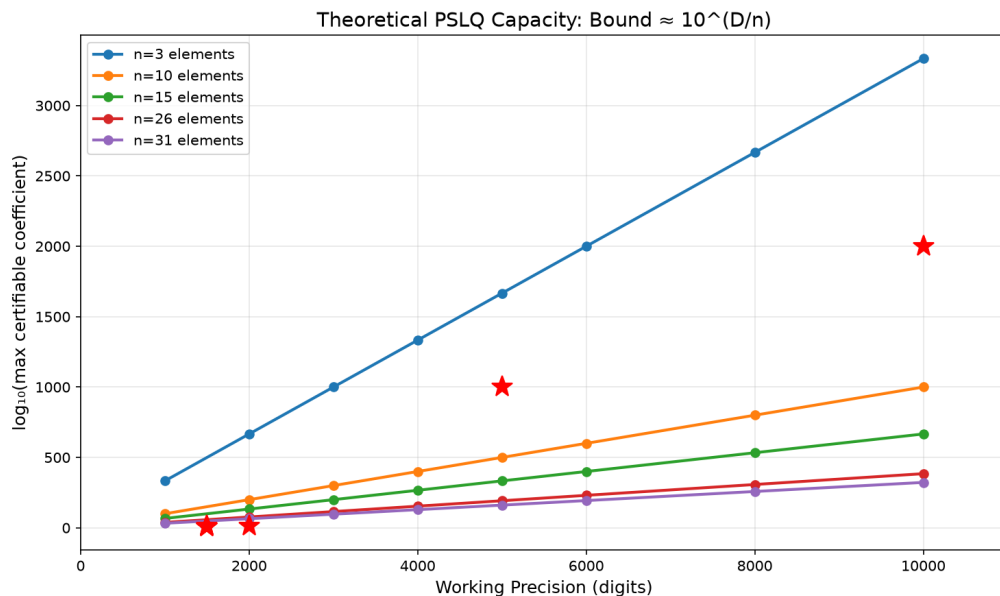
The internal norm bound of PSLQ for the main test $\{\zeta(3), \pi^2, 1\}$ at 10000 digits. The norm grows exponentially with each iteration until it exceeds $\text{maxcoeff} = 10^{2000}$ (red dashed line), at which point the algorithm certifies non-existence. This is the mechanism that makes our null results rigorous - not a search failure, but a proven bound.

Figure 2: Continued Fraction Partial Quotients



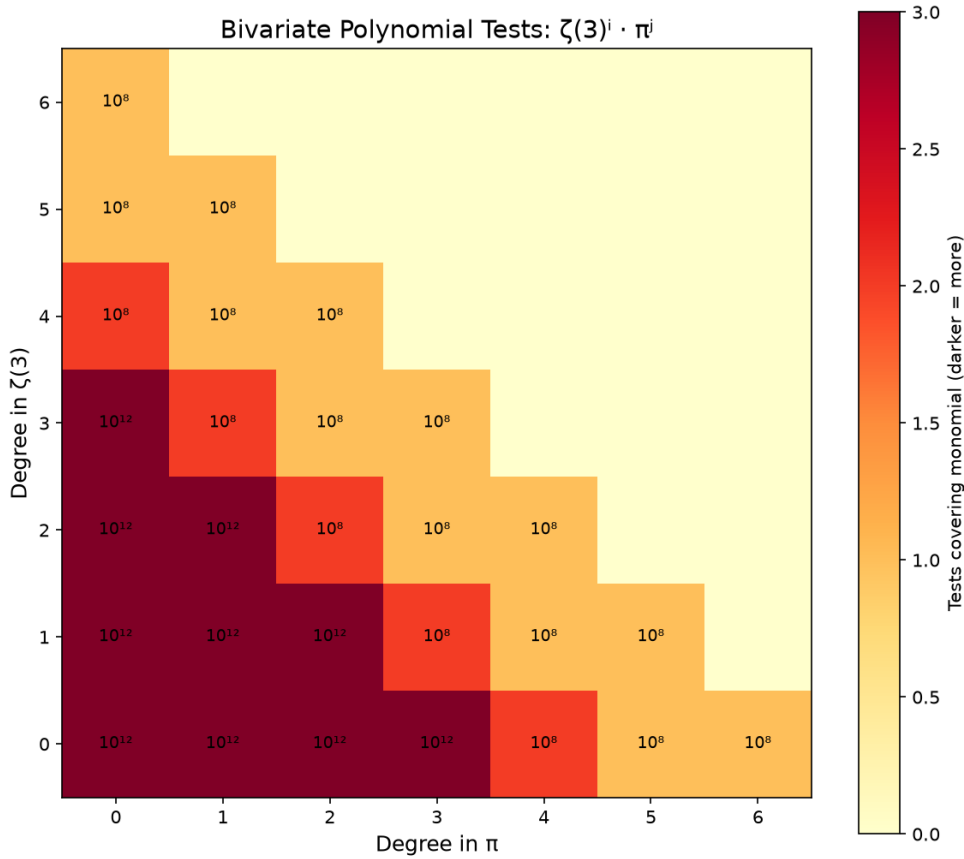
First 200 partial quotients of $\zeta(3)$ (top) and $\delta = \pi^2/8 - \zeta(3)$ (bottom). Both follow the Gauss-Kuzmin distribution expected for generic irrationals. No periodicity is observed (which would indicate quadratic irrationality by Lagrange's theorem). The red line marks Khinchin's constant $K \approx 2.69$.

Figure 3: Coefficient Bound vs Precision



Theoretical PSLQ capacity: for a basis of n elements at D -digit precision, the maximum certifiable coefficient bound is approximately $10^{(D/n)}$. Red stars mark our actual tests. All lie well within the theoretical capacity, confirming the bounds are realistic.

Figure 4: Bivariate Polynomial Test Coverage



Heatmap showing which monomials $\zeta(3)^i \cdot \pi^j$ were tested. Darker cells are covered by more independent tests. We tested all joint polynomials of total degree ≤ 6 : monomials of total degree ≤ 3 carry a certified coefficient bound of 10^{12} , and all others up to total degree 6 carry 10^8 .

5. Discussion

5.1 Interpretation

The central result (Result A) establishes that no relation $a \cdot \zeta(3) + b \cdot \pi^2 + c = 0$ exists with coefficients below 10^{2000} . Combined with the supporting results, this provides a consistent computational picture: no algebraic relation between $\zeta(3)$ and π was found within the tested bounds.

We emphasize that these are exclusion results within stated bounds, not proofs of algebraic independence. A relation with coefficients exceeding 10^{2000} could exist in principle. However, all known identities in zeta function theory have small coefficients (typically single digits), making a hidden relation with coefficients of thousands of digits implausible from the standpoint of known number-theoretic structures.

5.2 Comparison with Prior Work

Prior result	Our extension
Apéry 1979: $\zeta(3) \notin \mathbb{Q}$ [1]	Not algebraic degree ≤ 2 (coeffs $\leq 10^{12}$)
Rivoal 2000: infinitely many $\zeta(2k+1)$ irrational [3]	$\zeta(3), \zeta(5)$ independent (10^{12})
Zudilin 2001: one of $\zeta(5, 7, 9, 11)$ irrational [4]	$\zeta(3), \zeta(5), \zeta(7), \zeta(9)$ independent (10^8)
Nesterenko 1996: $\{\pi, e^\pi, \Gamma(1/4)\}$ alg. indep. [5]	$\zeta(3)$ independent from this triple (10^{12})
Bailey-Borwein 2005: experimental methods [6]	Linear bound 10^{2000} ; height bound 10^8 at degree 30

5.3 Limitations

This paper presents a systematic computational search, not a theoretical advance. Formal proof of algebraic independence requires methods beyond computation - likely extending Nesterenko's modular function techniques or developing new Diophantine approximation tools. Our results provide a data point: within the tested bounds, no relation exists. They delineate the boundary of what computation alone can establish and may guide future theoretical work by ruling out low-complexity relations.

Future non-algebraic tests include continued-fraction analysis of $\zeta(3)/\pi^3$, numerical verification of integral representations, and targeted series identities mixing $\zeta(3)$ and π .

6. Conclusion

No algebraic relation between $\zeta(3)$ and π was found within the tested bounds. Across 34 independent tests at precisions up to 10000 digits, every PSLQ computation returned a certified null result.

The three headline results: $\zeta(3) \neq (a/b) \cdot \pi^2 + c/d$ with coefficients up to 10^{2000} ; $\zeta(3)/\pi^3$ is not algebraic of degree ≤ 30 with height $\leq 10^8$; and $\zeta(3)$ and π satisfy no joint polynomial of total degree ≤ 6 with coefficients $\leq 10^8$. These bounds far exceed the coefficient sizes of every known identity among zeta values - for comparison, $\zeta(2) = \pi^2/6$ has coefficients 1 and 6.

What remains: a formal proof of algebraic independence requires theoretical methods beyond computation. Our results delineate what computation can currently certify: any algebraic relation between $\zeta(3)$ and π , if one exists, must have coefficients exceeding the certified bound of the corresponding test in the catalogue - in the linear case, coefficients larger than 10^{2000} - or degree beyond the tested range. The question remains open, but the computational evidence is now extensive and, for the first time in this line of work, uniformly certified.

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Appendix A: PSLQ Test Summary

All tests run on Apple M3 (8 cores). Times are for the PSLQ step only. Test numbers match Section 3.2.

Category 1: Linear independence from π

#	Basis	Size	Digits	Bound	Time	Result
1	$\{\zeta(3), \pi^2, 1\}$	3	10000	10^{2000}	120s	No relation
2	$\{\zeta(3), \pi^3, 1\}$	3	5000	10^{1000}	19.5s	No relation
3	$\{\zeta(3), \pi^2, \pi^4, 1\}$	4	2000	10^{10}	0.25s	No relation
4	$\{\zeta(3), \pi^2, \pi^4, \pi^6, 1\}$	5	1000	10^8	0.19s	No relation

#	Basis	Size	Digits	Bound	Time	Result
5	$\{\zeta(3), \pi^2, \pi^4, \pi^6, \pi^8, \pi^{10}, 1\}$	7	3000	10^8	3.9s	No relation

Category 2: Algebraicity of $\zeta(3)$

#	Basis	Size	Digits	Bound	Time	Result
6	$\{1, \zeta(3), \zeta(3)^2\}$	3	2000	10^{12}	0.07s	No relation
7	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3\}$	4	2000	10^{10}	0.26s	No relation
8	$\{1, \zeta(3), \zeta(3)^2, \zeta(3)^3, \zeta(3)^4\}$	5	2000	10^8	0.59s	No relation

Category 3: Algebraicity of $\zeta(3)/\pi^3$ and bivariate tests

#	Basis	Size	Digits	Bound	Time	Result
9	$\{(\zeta(3)/\pi^3)^k : k=0..10\}$	11	1500	10^{12}	15.0s	No relation
10	$\{(\zeta(3)/\pi^3)^k : k=0..15\}$	16	1500	10^9	60.5s	No relation
11	$\{(\zeta(3)/\pi^3)^k : k=0..25\}$	26	1500	10^{10}	568s	No relation
12	$\{(\zeta(3)/\pi^3)^k : k=0..30\}$	31	1500	10^8	1077s	No relation
13	$\{\zeta(3)^i \pi^j : i+j \leq 3\}$	10	2000	10^{12}	18.4s	No relation
14	$\{\zeta(3)^i \pi^j : i+j \leq 4\}$	15	2000	10^8	76.7s	No relation
15	$\{\zeta(3)^i \pi^j : i+j \leq 6\}$	28	1500	10^8	798s	No relation

Category 4: Odd zeta values and other constants

#	Basis	Size	Digits	Bound	Time	Result
16	$\{\zeta(3), \zeta(5), 1\}$	3	3000	10^{12}	0.09s	No relation
17	$\{\zeta(3), \zeta(5), \zeta(7), 1\}$	4	3000	10^{10}	0.38s	No relation
18	$\{\zeta(3), \zeta(5), \zeta(7), \zeta(9), 1\}$	5	1500	10^8	0.32s	No relation
19	$\{\zeta(3), \pi, e^\pi, \Gamma(1/4), 1\}$	5	4000	10^{12}	2.1s	No relation
20	$\{\zeta(3), \Gamma(1/4)^4/\pi^3, \pi^2, 1\}$	4	4000	10^{12}	0.88s	No relation
21	$\{\zeta(3), G, \pi^3, 1\}$	4	3000	10^{10}	0.41s	No relation
22	$\{\zeta(3), G^2, G \cdot \pi, \pi^2, G, 1\}$	6	3000	10^{10}	2.2s	No relation
23	$\{\zeta(3)^2, G^2, \zeta(3) \cdot G, \pi^4, \zeta(3), G, \pi^2, 1\}$	8	2000	10^8	4.3s	No relation
24	$\{\zeta(3)^2, \zeta(5) \cdot \pi, \zeta(7), \pi^6, \pi^4, 1\}$	6	3000	10^8	2.0s	No relation
25	$\{\zeta(3), \zeta(3) \cdot \pi^2, \zeta(5) \cdot \pi^2, \zeta(5), \pi^4, \pi^2, 1\}$	7	3000	10^8	4.0s	No relation

#	Basis	Size	Digits	Bound	Time	Result
26	$\{\zeta(3)^2, \zeta(5), \pi^6, \pi^4, \pi^2, 1\}$	6	3000	10^{10}	2.3s	No relation
27	$\{\zeta(3), \zeta(2)\zeta(3)-\zeta(5), \zeta(5), \pi^2, 1\}$	5	1000	10^8	0.19s	No relation

Category 5: L-values and BBP-type tests

#	Basis	Size	Digits	Bound	Time	Result
28	$\{\zeta(3), L(E_{32}, 2), \pi^2, 1\}$	4	2000	10^{10}	0.24s	No relation
29	$\{\zeta(3), L(\chi_{4,3}), \pi^2, 1\}$	4	2000	10^{10}	0.24s	No relation
30	$\{\zeta(3), L(E_{32}, 2), L(\chi_{4,3}), \pi^2, 1\}$	5	2000	10^8	0.55s	No relation
31	$\{\zeta(3), \pi^2 \ln 2, \ln^3 2, \ln^2 2, \ln 2, \pi^2, 1\}$	7	4000	10^{11}	9.2s	No relation
32	$\{\zeta(3), Li_3(1/3), \pi^2 \ln 3, \ln^3 3, 1\}$	5	2000	10^{10}	0.59s	No relation
33	$\{\zeta(3), Li_3(1/4), \pi^2 \ln 2, \ln^3 2, 1\}$	5	2000	10^{10}	0.66s	No relation

Category 6: Multiple zeta values

#	Basis	Size	Digits	Bound	Time	Result
34	$\{\zeta(3), \zeta(3, 2), \zeta(2, 3), \pi^5, 1\}$	5	4000	10^{10}	1.6s	No relation

Category 7: Validation (known identities recovered)

#	Basis	Size	Digits	Bound	Time	Result
35	$\{\zeta(3), Li_3(1/2), \pi^2 \ln 2, \ln^3 2\}$	4	5000	10^6	0.09s	FOUND [21,-24,-2,4]
36	$\{\zeta(3), Li_3(-1)\}$	2	5000	10^6	<0.01s	FOUND [3, 4]
37	$\{\zeta(3)^2, \zeta(6), \pi^6, 1\}$	4	5000	10^6	0.07s	FOUND [0,-945,1,0]

Appendix B: Continued Fraction Data

Continued fraction of $\delta = \pi^2/8 - \zeta(3)$, first 100 partial quotients:

[0; 31, 1, 1, 1, 1, 1, 20, 4, 3, 1, 1, 11, 1, 4, 23, 9, 39, 1, 1, 6, 1, 1, 35, 1, 7, 6, 1, 2, 2, 1, 1, 1, 1, 5, 1, 1, 11, 1, 2, 6, 1, 5, 23, 2, 2, 2, 7, 2, 6, 155, 2, 1, 1, 59, 1, 3, 1, 16, 9, 1, 3, 1, 1, 1, 4, 13, 5, 1, 4, 4, 4, 1, 3, 1, 3, 3, 1, 3, 1, 4, 3, 4, 11, 2, 1, 1, 2, 18, 7, 1, 1, 15, 2, 1, 2, 71, 1, 4, 1, 1, 8]

Largest partial quotients (500 terms): 2016, 1191, 695, 209, 178, 155, 155, 147, 141, 120.

Appendix C: Computational Reproducibility

The complete set of tests can be reproduced by running `run_tests.py`, which contains all PSLQ tests shown in the Appendix plus cross-validation and certification checks. Every result reported in this paper is generated by that script.

The main result ($\{\zeta(3), \pi^2, 1\}$ at 10000 digits with bound 10^{2000}) can be reproduced with the following code:

```
from mpmath import mp, zeta, pi, pslq

mp.dps = 10000
z3 = zeta(3)
result = pslq([z3, pi**2, mp.mpf(1)], maxcoeff=10**2000, maxsteps=20000)
print(result is None) # True means no relation found
```

The `maxsteps` argument is essential: at the default iteration limit PSLQ returns None without certifying anything (Section 2.1). The norm bound crosses 10^{2000} at iteration 10,687, in roughly two minutes. The full suite takes approximately 55 minutes on an Apple M3 processor, dominated by the degree-30 algebraicity certificate at 1500 digits.

License

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